ARPIT SINGH

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SUMMARY

Third-year Computer Engineering student seeking a software development internship. Experienced in building full-stack web applications with React, Node.js, and PostgreSQL, as well as Python-based tools and AI-integrated applications using REST APIs and the Google Gemini API.

EDUCATION

Vidyalankar Institute of Technology, Mumbai Expected 2027

Bachelor of Engineering in Computer Engineering

Relevant Coursework: Computer Networks, Computer Graphics, Software Engineering, Theory of Computer Science

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, HTML, CSS
Frontend:	React, Vite, Next.js, Tailwind CSS
Backend:	Node.js, Express, FastAPI, REST APIs, WebSockets, WebRTC, JWT Authentication, Prisma ORM
Databases:	PostgreSQL, Supabase, Data Warehousing (Star Schema)
APIs & Integrations:	Google Gemini API, Twilio (WhatsApp), GitHub REST API, OpenF1 API, Open-Meteo
Tools & Platforms:	Git, GitHub, Docker, Chrome Extensions (Manifest V3), Vercel, Render, WSL

PROJECTS

MeshStream: Peer-Assisted Live Video Streaming | *TypeScript, WebRTC, WebSockets, hls.js, MediaMTX, Node.js*

- Built a hybrid peer-to-peer live streaming platform: broadcasters publish via WHIP into MediaMTX, which serves Low-Latency HLS, and viewers exchange segments over WebRTC DataChannels coordinated by a WebSocket tracker.
- Implemented deadline-based peer fetching with HTTP fallback via a custom hls.js loader, chunked transfer with backpressure, bounded LRU caches, and telemetry that reports P2P offload ratio.

LeetCode FSRS Tracker | *Next.js, TypeScript, PostgreSQL (Supabase), Prisma, NextAuth.js, Tailwind CSS*

- Built and deployed on Vercel a spaced-repetition web app that schedules LeetCode reviews with FSRS-6 (ts-fsrs), targeting 90% recall with a paced daily queue ordered by lowest retrievability.
- Added LeetCode sync, CSV import, per-pattern retention tracking, and NextAuth.js authentication.

RegimeTrader: Regime-Aware Stock Direction Predictor | *Python, SQL, Streamlit, XGBoost*

- Built a star-schema data warehouse of Nifty 50 daily history with ETL, OLAP queries, regime detection, and classifiers (logistic regression, decision tree, random forest, XGBoost) for a Data Warehousing and Mining course project.
- Developed a backtesting and paper-trading engine, evaluated on a held-out 2024-2026 test period against buy-and-hold, equal-weight, and moving-average baselines with confidence intervals; 500+ automated tests.

Task & Project Manager | *React, Vite, Node.js, Express, PostgreSQL, Prisma, JWT*

- Built a full-stack task and project management application with complete CRUD functionality using a React (Vite) frontend and an Express REST API.
- Modeled and queried relational data in PostgreSQL using Prisma ORM.
- Implemented JWT authentication with httpOnly cookies to reduce exposure to XSS-based token theft.
- Deployed using Render and Vercel.

Krishi-Mitra: WhatsApp Advisory Bot for Farmers | *Python, FastAPI, Twilio, PostgreSQL, Gemini API*

- Developed a WhatsApp chatbot backend with FastAPI and Twilio webhooks that delivers AI-generated farming advisories using the Google Gemini API.
- Integrated Open-Meteo weather data and PostgreSQL storage; supports English, Hindi, Marathi, and Punjabi.
- Added voice-note transcription (speech_recognition, pydub) and voice replies (gTTS).

Deepguard: Deepfake Detection Tool | *Python, DeepfakeBench, EfficientNet, Xception*

- Building a deepfake detection web tool on the DeepfakeBench framework that compares multiple detection models and surfaces model disagreement in the interface.
- Designed a per-frame evidence view and uncertainty-aware verdicts after identifying reliability gaps in an earlier version.

LeetCode-to-GitHub Sync Chrome Extension | *JavaScript, Chrome Extension (MV3), GitHub API*

- Built a Chrome extension that syncs LeetCode submissions to GitHub while preserving every submission per problem instead of overwriting earlier solutions.
- Organizes commits by problem name and by topic using the GitHub Contents API.

OpenF1 Terminal Dashboard | *Python, Textual, Docker, REST API*

- Developed a terminal dashboard for race weekends showing lap times, live driver positions, standings, and tire compounds from the OpenF1 API.
- Containerized the application with Docker.

ADDITIONAL

- Maintain a public GitHub repository of LeetCode solutions: github.com/arp1tSingh/LeetCode